Task 10 - Compare Expectations with Execution

C Programming Assignment - Rural Electricity Billing System

Large totals

4,000 bills times ■60,000 = ■240,000,000 (■24 crore). 60,000 bills times ■60,000 = ■3,600,000,000 (■360 crore). 800,000 bills times ■60,000 = ■48,000,000,000 (■4,800 crore).

Overflow comparison

A typical 32-bit signed int has maximum 2,147,483,647, so it cannot hold ■48,000,000,000 in rupees. A signed 64-bit integer can hold it. Check actual limits on the target platform.

Float vs integer

Integer paise gives exact whole-paise arithmetic until overflow. Float and double can introduce rounding errors. Use integer paise for money and check additions and multiplications.

Type sizes

C does not guarantee identical sizes for int, long and long long. Use sizeof and limits.h. Use int64_t/uint64_t and PRI-format macros where available.

Complexity

Each numeric operation is O(1) time and space in the algorithmic model; actual performance depends on hardware and compiler.

Important: illustrative tariff rates are used because the screenshot does not show the complete official tariff table. Replace them with your instructor's prescribed rates.